

Industrial Internship Report on " FoodDash-Express: AI-Powered Full-Stack Food Delivery Platform"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "FoodDash-Express," an AI-powered full-stack food delivery web application inspired by Swiggy and Zomato. It enables customers to order food online, restaurants to manage menus and orders, delivery partners to track deliveries, and administrators to monitor the platform. The application also includes an AI-powered food recommendation system (Rasoi AI), simulated UPI payments, live order tracking, and analytics dashboards.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

This internship provided me with an excellent opportunity to gain practical knowledge and hands-on experience in **Full-Stack Web Development**. Over the course of **six weeks**, I learned modern web technologies, software development practices, and the complete process of designing and developing a real-world application. The internship helped me strengthen my technical skills while improving my problem-solving, analytical thinking, teamwork, and project management abilities.

As part of the internship, I developed **FoodDash-Express**, an AI-powered full-stack food delivery web application inspired by leading platforms like Swiggy and Zomato. The project enables customers to order food online, restaurants to manage menus and orders, delivery partners to handle deliveries, and administrators to monitor platform activities. It also integrates **Rasoi AI**, powered by Google Gemini, to provide personalized food recommendations, making the application more interactive and user-friendly.

The internship organized by **Upskill Campus (USC)** in collaboration with **UniConverge Technologies (UCT)** provided a well-structured learning environment with study materials, project guidance, and weekly assignments. The program was planned systematically, beginning with the fundamentals of programming and full-stack development, followed by project planning, implementation, testing, and documentation. This step-by-step approach helped me successfully complete the project while continuously enhancing my technical knowledge.

Throughout the internship, I gained valuable experience in **React, TypeScript, Tailwind CSS, Node.js, Express.js, REST APIs, AI integration, responsive web design, and software architecture**. I also learned the importance of writing clean code, debugging applications, and developing scalable web solutions.

I sincerely express my gratitude to **Upskill Campus (USC)** and **UniConverge Technologies (UCT)** for providing this valuable internship opportunity and learning platform. I would also like to thank my mentors, trainers, project coordinators, and everyone who supported and guided me throughout the internship. Their encouragement and valuable feedback played an important role in my learning journey.

Finally, I encourage my juniors and peers to make the best use of such internship opportunities. Stay curious, practice consistently, work on real-world projects, and never stop learning. The knowledge and experience gained through practical implementation are invaluable for building a successful career in the software industry.

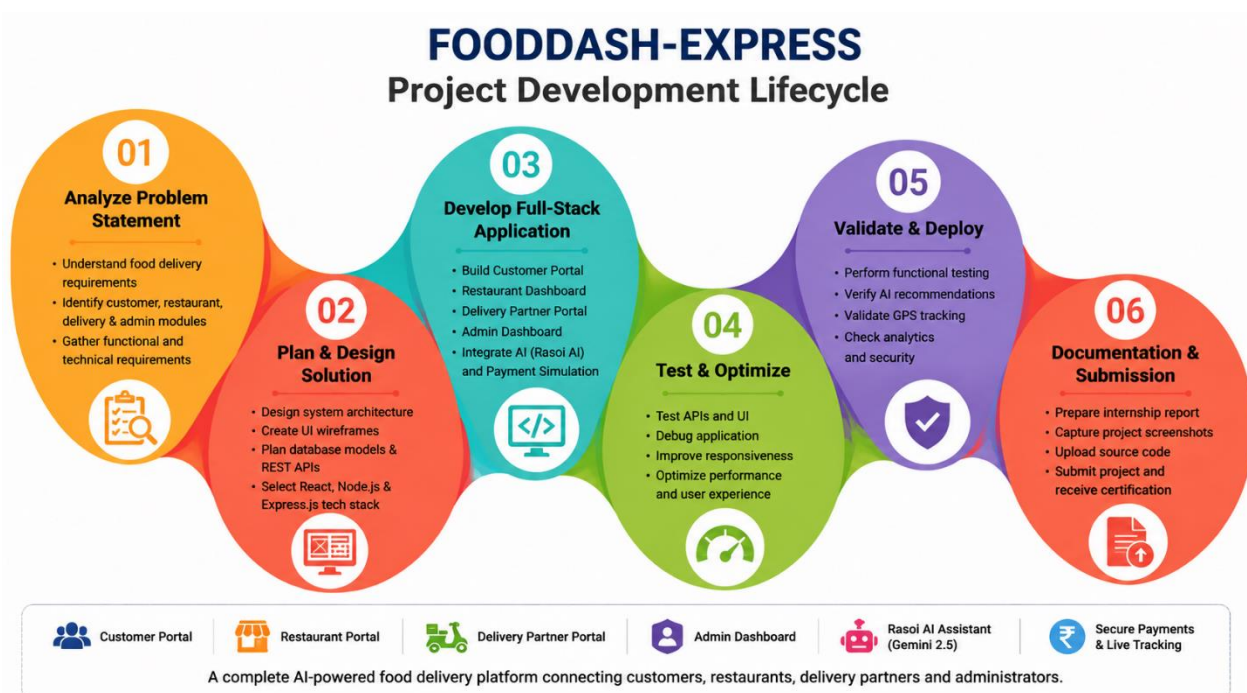


Fig 1.1 : FoodDash-Express Project Development Lifecycle

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



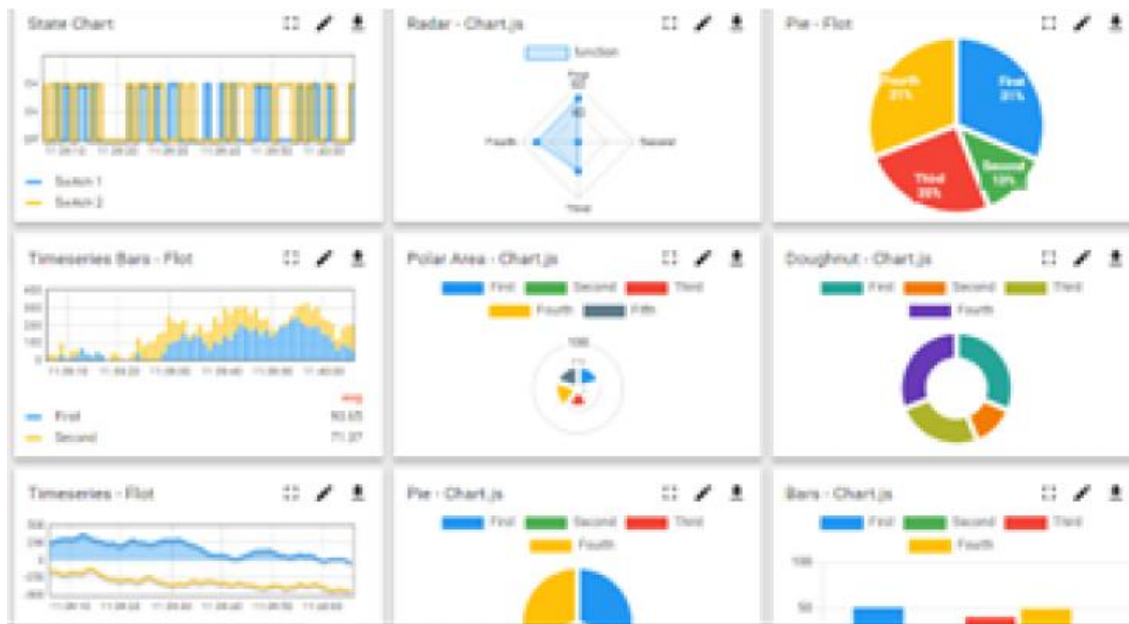
i. UCT IoT Platform ()

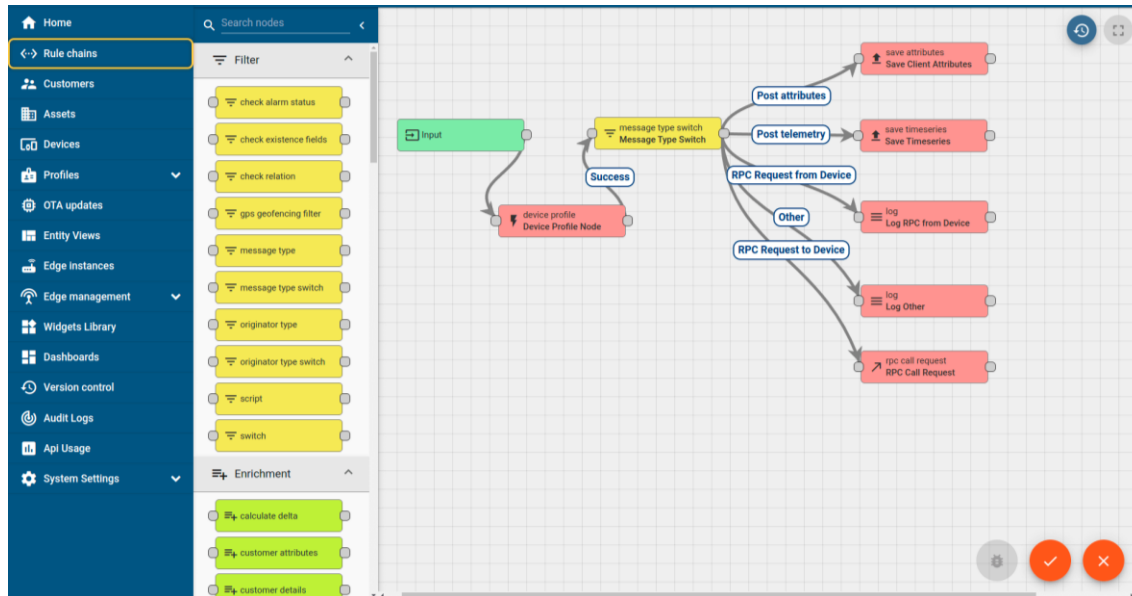
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





FACTORY WATCH

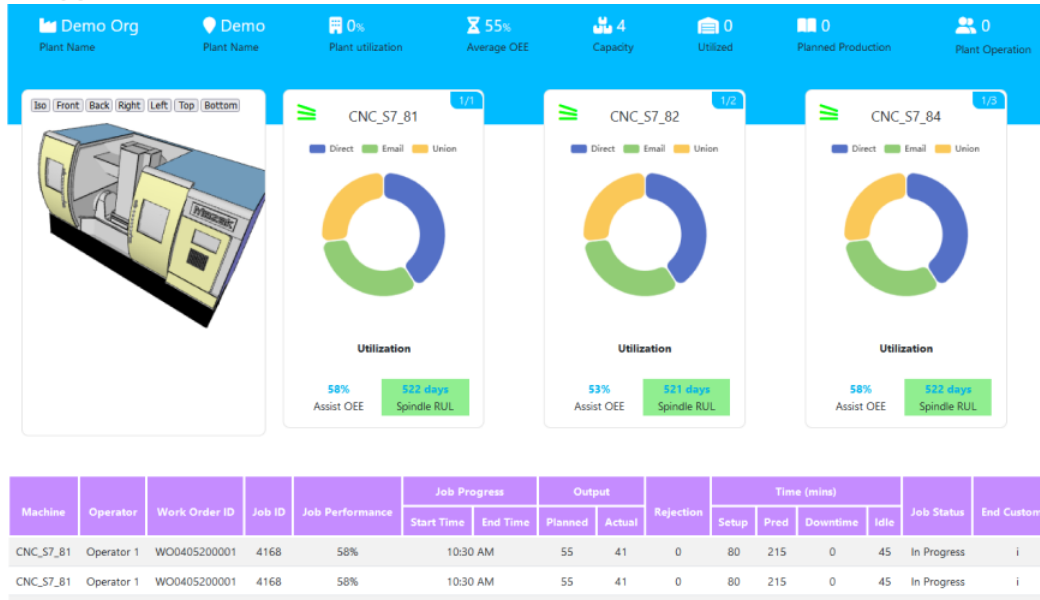
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



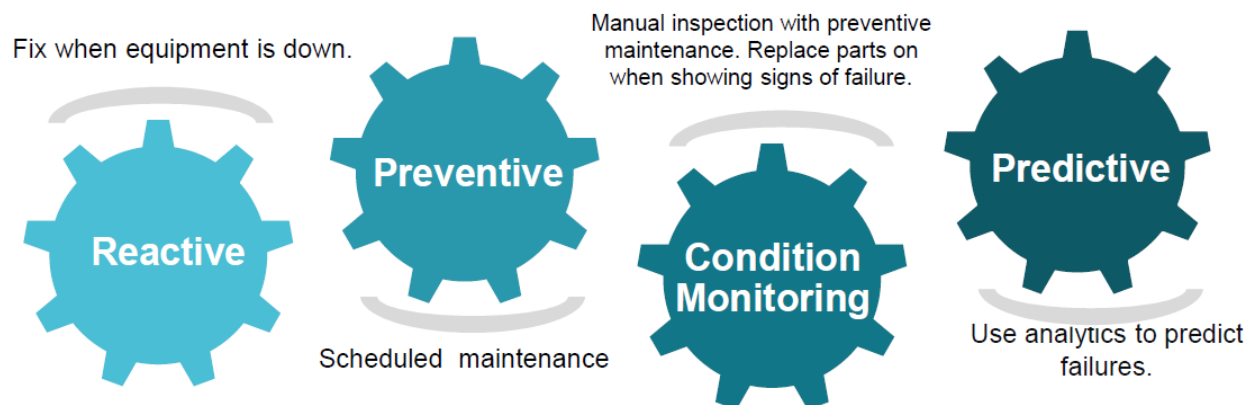


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

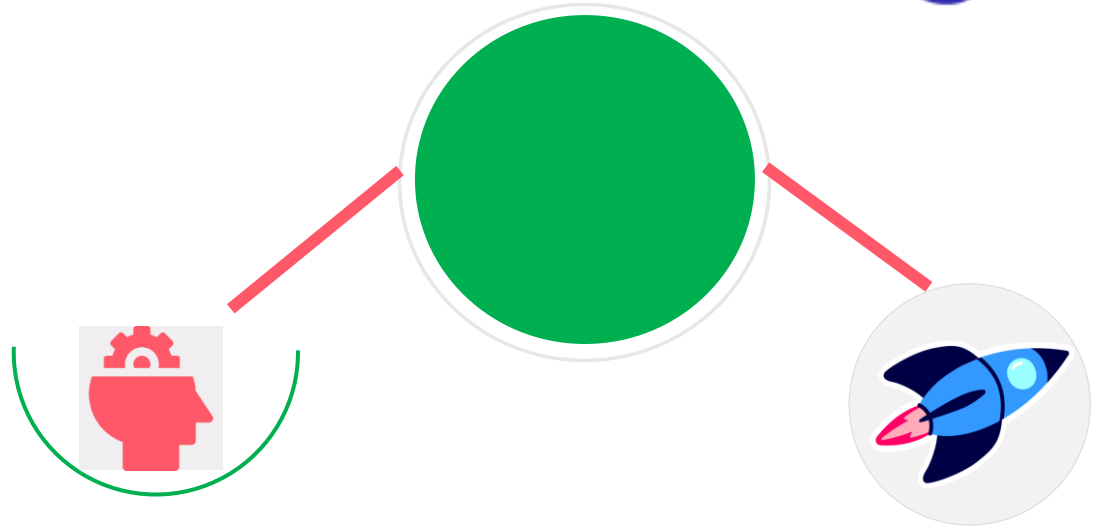
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

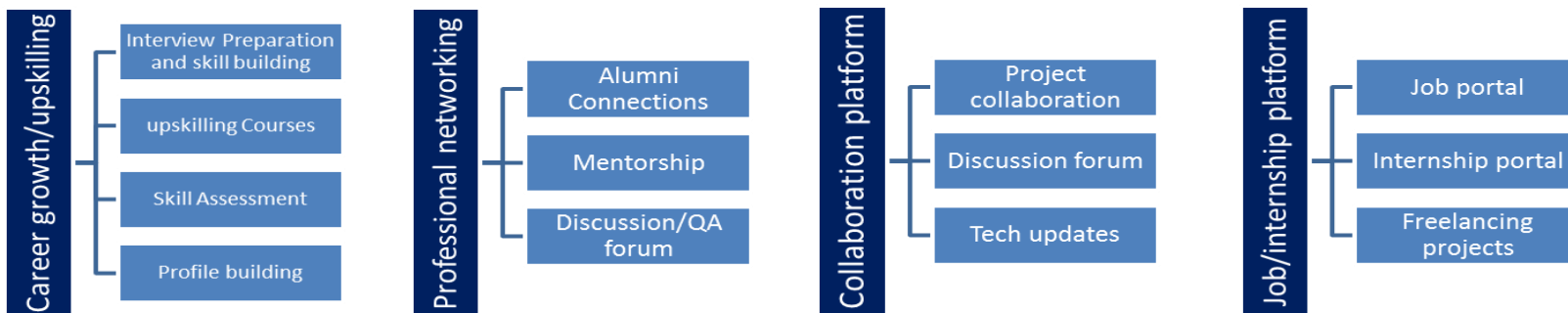
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objectives of this internship program were to:

- Gain practical experience in Full-Stack Web Development.
- Understand the software development life cycle (SDLC).
- Develop a real-world web application using modern technologies.
- Improve programming, debugging, and problem-solving skills.
- Learn REST API development and frontend-backend integration.
- Gain exposure to AI integration in web applications.
- Enhance teamwork, communication, and project management skills.

2.5 Reference

[1] React Official Documentation – <https://react.dev>

[2] Express.js Documentation – <https://expressjs.com>

[3] Node.js Documentation – <https://nodejs.org>

[4] Google Gemini AI Documentation – <https://ai.google.dev>

[5] Tailwind CSS Documentation – <https://tailwindcss.com>

2.6 Glossary

Terms	Acronym
Artificial Intelligence	AI
Application Programming Interface	API
Representational State Transfer	REST
Hypertext Transfer Protocol	HTTP
JavaScript Object Notation	JSON
User Interface	UI
User Experience	UX
Global Positioning System	GPS
Unified Payments Interface	UPI
Full Stack Development	FSD
Create, Read, Update, Delete	CRUD
Google Gemini	GenAI
Node Package Manager	NPM
HyperText Markup Language	HTML
Cascading Style Sheets	CSS

3 Problem Statement

The traditional food ordering process often involves difficulties for customers in discovering nearby restaurants, checking available food items, comparing prices, and tracking order delivery status. Restaurants also face challenges in managing menus, handling customer orders, updating food availability, and coordinating with delivery partners efficiently.

The objective of this project is to develop a **full-stack food delivery application** that creates a seamless connection between customers, restaurants, and delivery partners. The system allows restaurant owners to register, manage their menus, update prices, and process incoming orders. Customers can browse restaurants, search food items, add items to a unique cart, make payments, and track their orders in real time.

The application also focuses on improving user experience through features such as personalized food recommendations using AI, cuisine-based browsing, ratings and reviews, GPS-based delivery tracking, dynamic ETA updates, digital payment support, and role-based dashboards for customers, restaurants, delivery partners, and administrators.

The proposed solution, **FoodDash-Express**, aims to provide an efficient, scalable, and user-friendly digital platform that simplifies food ordering, restaurant management, and delivery operations while enhancing customer satisfaction.

4. Existing and Proposed solution

4.1 Existing Solutions

Currently, several food delivery platforms such as **Swiggy, Zomato, Uber Eats, and DoorDash** provide online food ordering services by connecting customers with restaurants. These platforms allow users to search restaurants, browse menus, place orders, make payments, and track deliveries. Restaurants can manage their food items, receive orders, and coordinate with delivery partners through dedicated dashboards.

However, existing solutions have some limitations:

- **Limited personalization:** Food recommendations are mostly based on previous orders, ratings, and popular items, with limited AI-based suggestions based on user preferences, budget, or dietary requirements.
- **Complex user experience:** Customers may need to navigate through multiple sections for searching, ordering, payment, and tracking.
- **Limited restaurant control:** Small restaurants may face difficulties in managing menus, promotions, analytics, and customer engagement effectively.
- **Generic delivery tracking:** Some platforms provide basic tracking but may lack detailed driver movement visualization and dynamic ETA updates.
- **High platform dependency:** Restaurants often depend heavily on third-party platforms, which may involve high commission charges and limited customization.
- **Limited integrated features:** Features such as AI food assistance, smart recommendations, loyalty programs, and detailed analytics are not always available in a single platform.

4.2 Proposed Solution

The proposed solution is **FoodDash-Express**, a full-stack Indian food delivery application designed to create an efficient connection between customers, restaurants, delivery partners, and administrators.

The system provides four role-based portals:

1. Customer Portal

- Browse restaurants and food items based on cuisine, ratings, and preferences.
- Search and filter dishes with Veg/Non-Veg classification.
- Add items to a unique cart and complete checkout.
- Support online payments (UPI simulation, card, and cash on delivery).
- Track delivery partner location with live GPS simulation and dynamic ETA updates.
- Save favorite restaurants and previous orders.

1. Restaurant Portal

- Register and manage restaurant profiles.
- Add, update, and remove food items.
- Manage prices, availability, discounts, and categories.
- Accept customer orders and update preparation status.
- View sales analytics and order information.

2. Delivery Partner Portal

- Manage delivery requests.
- Accept assigned orders.
- Update pickup and delivery status.
- Track delivery routes and earnings.

3. Admin Portal

- Monitor platform activities.
- View revenue, order statistics, and restaurant information.
- Manage coupons, categories, and system analytics.

4.3 Value Addition

The proposed FoodDash-Express application provides the following value additions:

- **AI-Powered Food Assistant (Rasoi AI):** Uses Google Gemini AI to provide personalized food recommendations based on user preferences, budget, and dietary choices.
- **Real-Time Delivery Tracking:** Provides live delivery partner tracking with dynamic estimated delivery time.
- **Indian Food Localization:** Supports Indian Rupee (₹), FSSAI-style Veg/Non-Veg indicators, Indian cuisines, and UPI payment simulation.
- **Role-Based Management System:** Provides separate dashboards for customers, restaurants, delivery partners, and administrators.
- **Smart Recommendation System:** Helps users discover suitable dishes through personalized suggestions.
- **Advanced Analytics:** Provides restaurant and admin insights through revenue trends, order statistics, and customer data.
- **Improved Restaurant Management:** Enables small and medium restaurants to efficiently manage menus, orders, pricing, and customer engagement through a single platform.

4.3 Code submission (Github link)

<https://github.com/deekshagowda123/FoodDash-Express.git>

4.4 Report submission (Github link) : first make placeholder, copy the link.

Report Link: (To be updated after uploading the project report to GitHub)

5. Proposed Design/ Model

5.1 System Design Overview

The proposed system, **FoodDash-Express**, is a full-stack food delivery application designed to connect customers, restaurants, delivery partners, and administrators through a single digital platform. The system follows a role-based architecture where each user type has dedicated functionalities and workflows.

The application starts with user registration and authentication, followed by different operational flows for customers, restaurants, delivery partners, and administrators. The system uses a React-based frontend, Express.js backend, and integrated AI services to provide a seamless food ordering and delivery experience.

5.1.1 Design Flow of the Proposed System

Step 1: User Registration and Role Selection

The system begins with user registration where users can access the platform based on their roles:

- **Customer:** Browse restaurants, order food, make payments, and track deliveries.
- **Restaurant Owner:** Manage restaurant details, food menus, pricing, availability, and incoming orders.
- **Delivery Partner:** Accept delivery requests, update delivery status, and complete deliveries.
- **Admin:** Monitor platform activities, analytics, users, and system performance.

Step 2: Restaurant and Menu Management

Restaurant owners can create and manage their restaurant profiles by:

- Adding restaurant information such as name, cuisine, address, and delivery details.
- Uploading food items with descriptions, images, prices, categories, and Veg/Non-Veg classification.
- Updating item availability, discounts, and menu changes.

The restaurant data is stored and managed through backend APIs.

Step 3: Customer Food Browsing and Selection

Customers can:

- Search and explore nearby restaurants.
- Filter food items based on cuisine, ratings, and preferences.
- View detailed food information including price, category, and availability.
- Receive personalized food recommendations using **Rasoi AI Assistant powered by Google Gemini**.
- Add selected items to a unique shopping cart.

Step 4: Cart and Order Processing

The cart module manages selected food items and calculates:

- Item subtotal.
- GST charges.
- Discounts and coupons.
- Final order amount.

After checkout, customers can select payment methods such as:

- UPI Payment Simulation.
- Credit/Debit Card.
- Cash on Delivery (COD).

The order details are sent to the selected restaurant through REST API communication.

Step 5: Restaurant Order Confirmation

The restaurant receives customer orders and performs the following actions:

- Accept or reject orders.
- Confirm preparation time.
- Update order status.

The order workflow follows:

ORDER_RECEIVED → PREPARING → READY → PICKED_UP → OUT_FOR_DELIVERY → DELIVERED

Step 6: Delivery Partner Assignment and Tracking

Once the food is prepared:

- A delivery partner is assigned to collect the order.
- The delivery partner updates pickup and delivery status.
- The customer receives live delivery updates.
- GPS-based tracking displays delivery partner movement and estimated delivery time.

Step 7: AI Recommendation and Analytics

Rasoi AI Chef Assistant

- Suggests dishes based on budget, dietary preferences, and user requirements.
- Provides quick food recommendations with add-to-cart options.

Analytics Module

- Tracks restaurant sales.
- Displays order statistics.
- Provides revenue trends and platform insights for administrators.

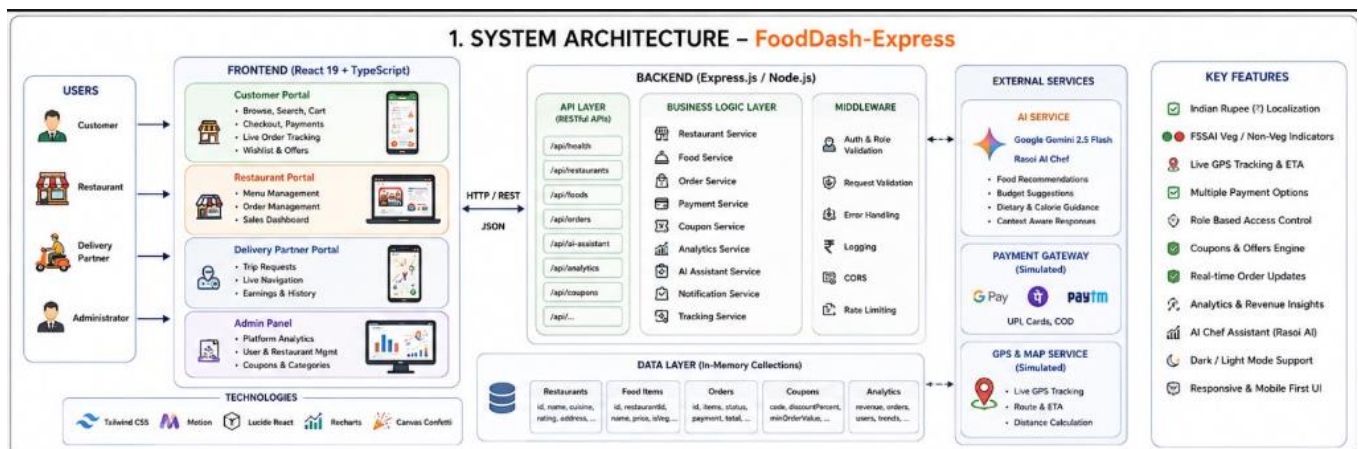


Figure 1: SYSTEM ARCHITECTURE DIAGRAM OF THE SYSTEM

5.2 High Level Diagram

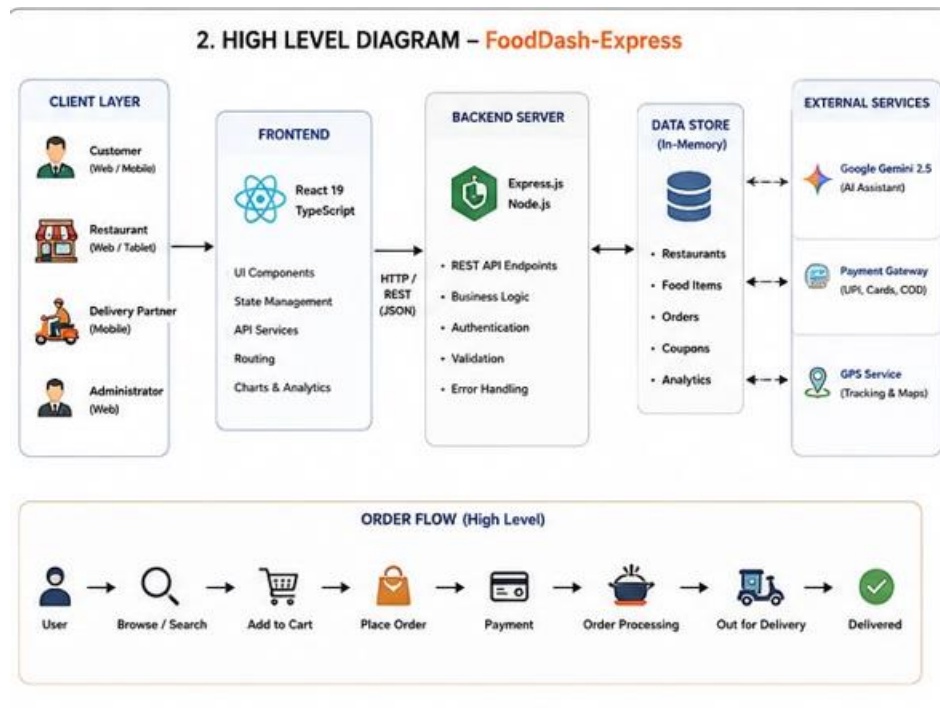


Figure 2: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.3 Low Level Diagram

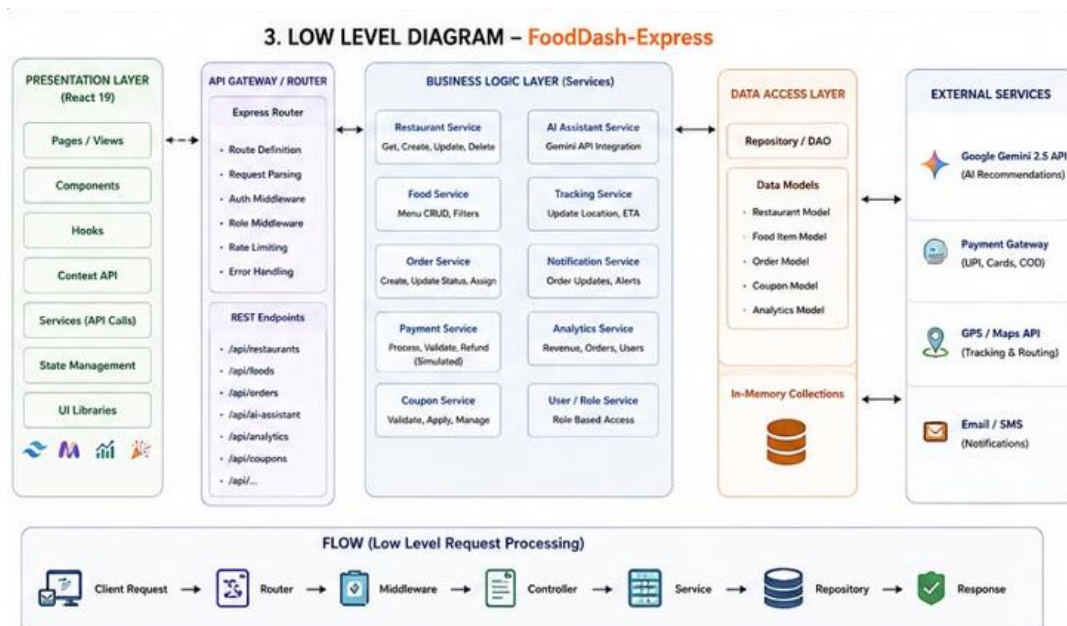


Figure 3: LOW LEVEL DIAGRAM OF THE SYSTEM

5.4 Interfaces

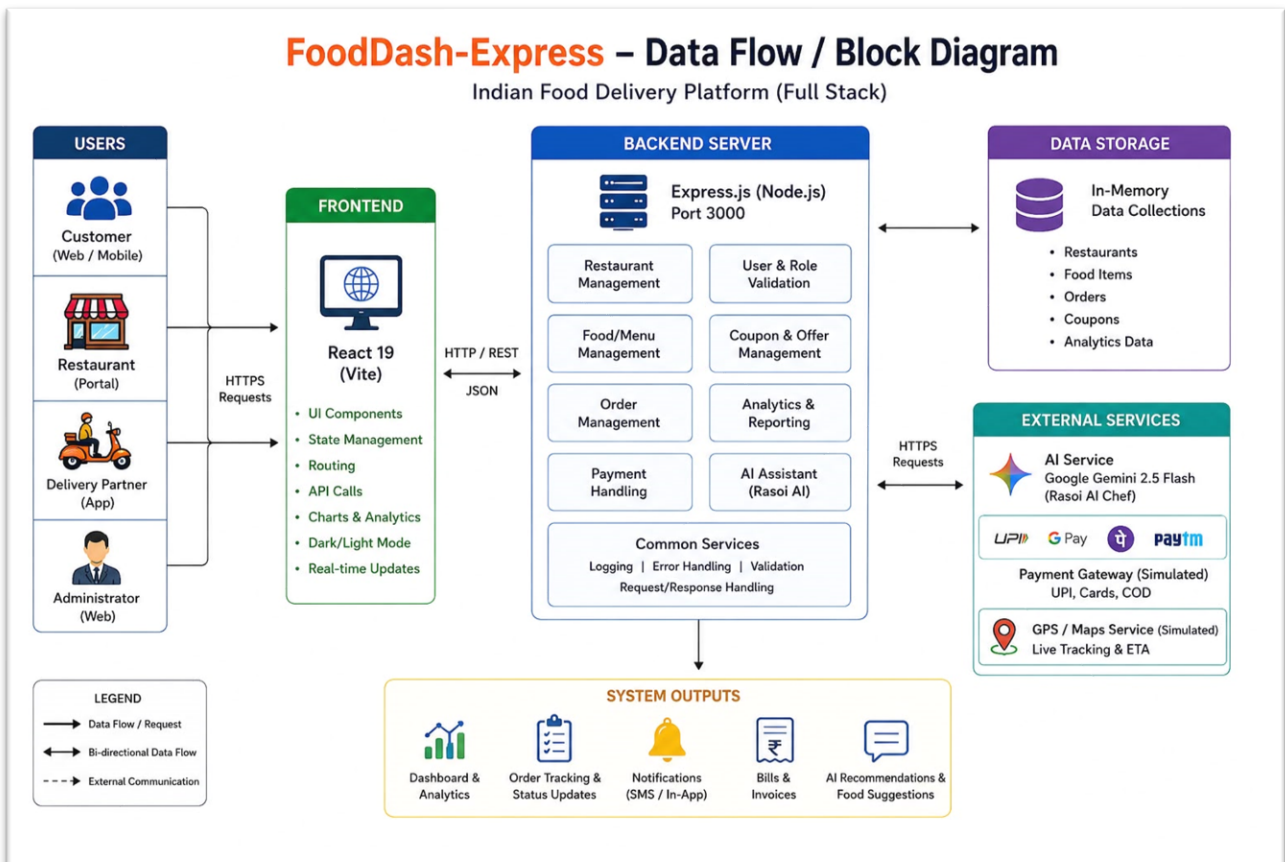


Figure 4: DATA FLOW DIAGRAM OF THE SYSTEM

6. Performance Test

The performance testing of the FoodDash-Express platform was conducted to evaluate the responsiveness, reliability, and efficiency of the application under normal operating conditions. The application was tested for page loading, REST API response time, order processing, AI response, and memory utilization.

6.1 Test Plan/ Test Cases

Test ID	Test Case	Expected Result	Status
PT-01	Load Home Page	Home page loads within 2 seconds	Pass
PT-02	Search Restaurants	Results displayed within 1 second	Pass
PT-03	View Menu Items	Menu loads correctly	Pass
PT-04	Add Item to Cart	Cart updates instantly	Pass
PT-05	Place Order	Order created successfully	Pass
PT-06	Update Order Status	Status changes without delay	Pass
PT-07	Rasoi AI Recommendation	AI response generated within 3–5 seconds	Pass
PT-08	Admin Dashboard	Analytics displayed correctly	Pass

6.2 Test Procedure

The following procedure was used:

1. Start the Express.js backend server.
2. Launch the React frontend using Vite.
3. Open the application in a web browser.
4. Perform restaurant search and menu browsing.
5. Add multiple food items to the cart.

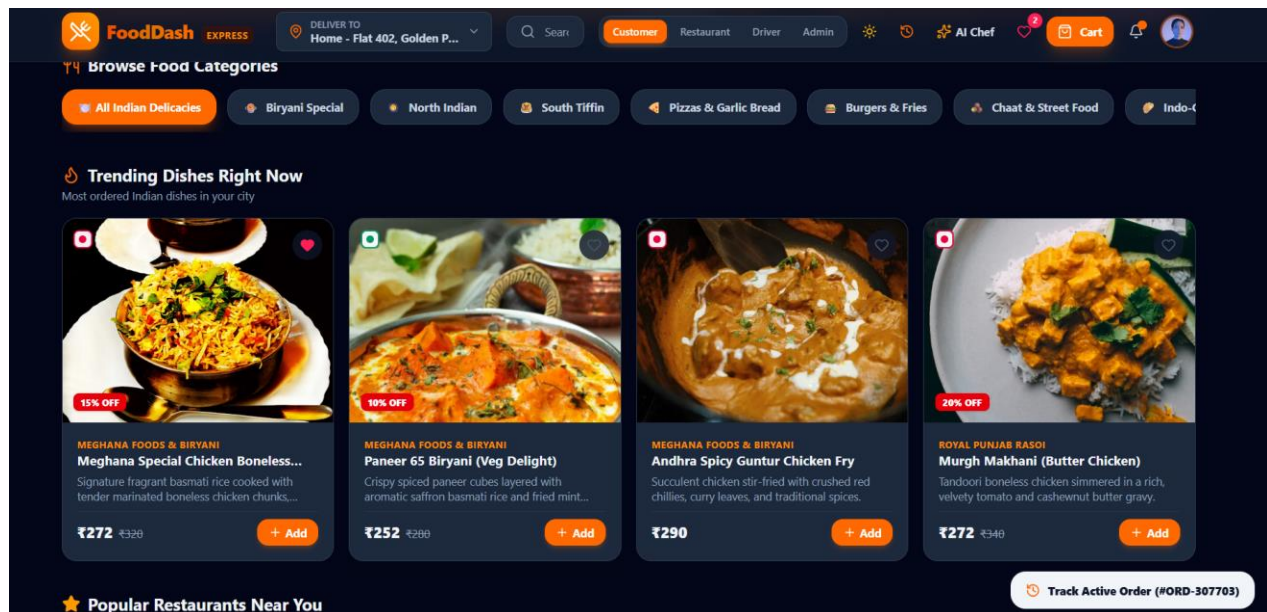
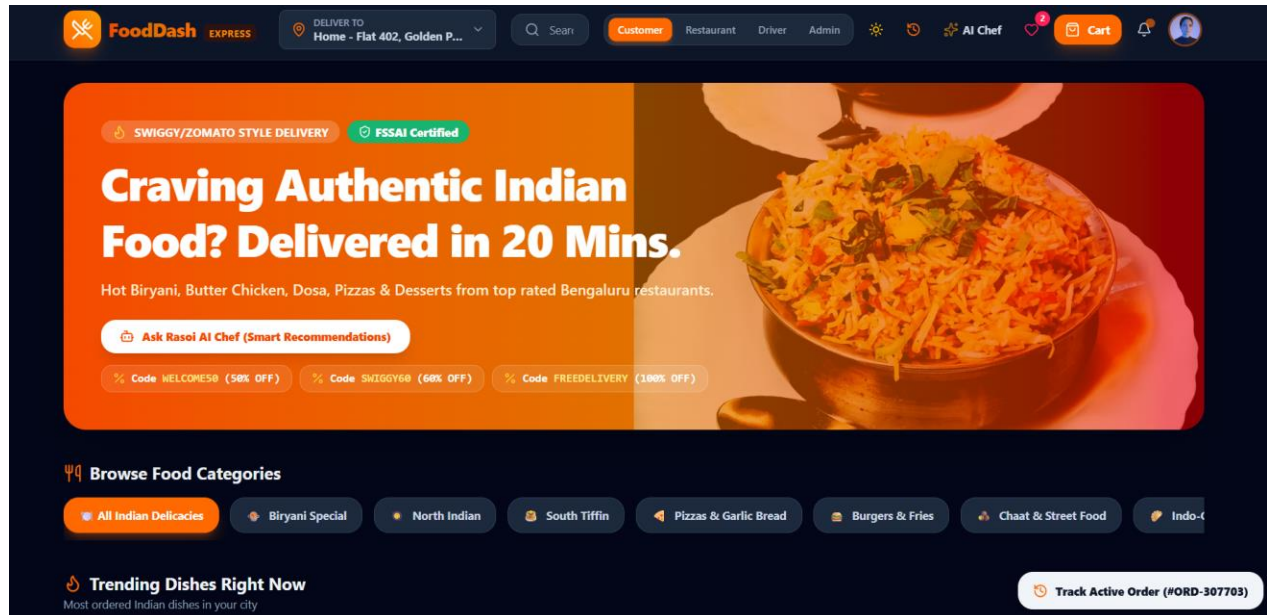
6. Place an order using the simulated payment gateway.
7. Track the order through all delivery stages.
8. Test the AI Chef Assistant with different food recommendation queries.
9. Measure response time and observe CPU and memory usage during execution.

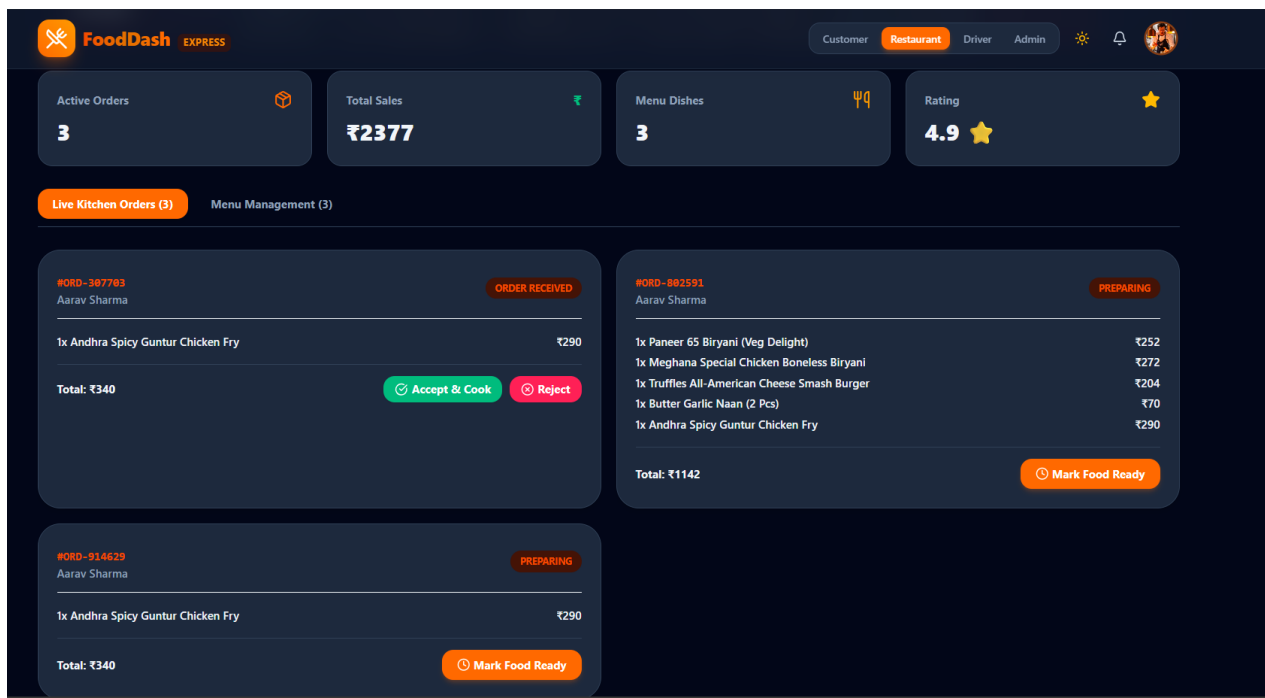
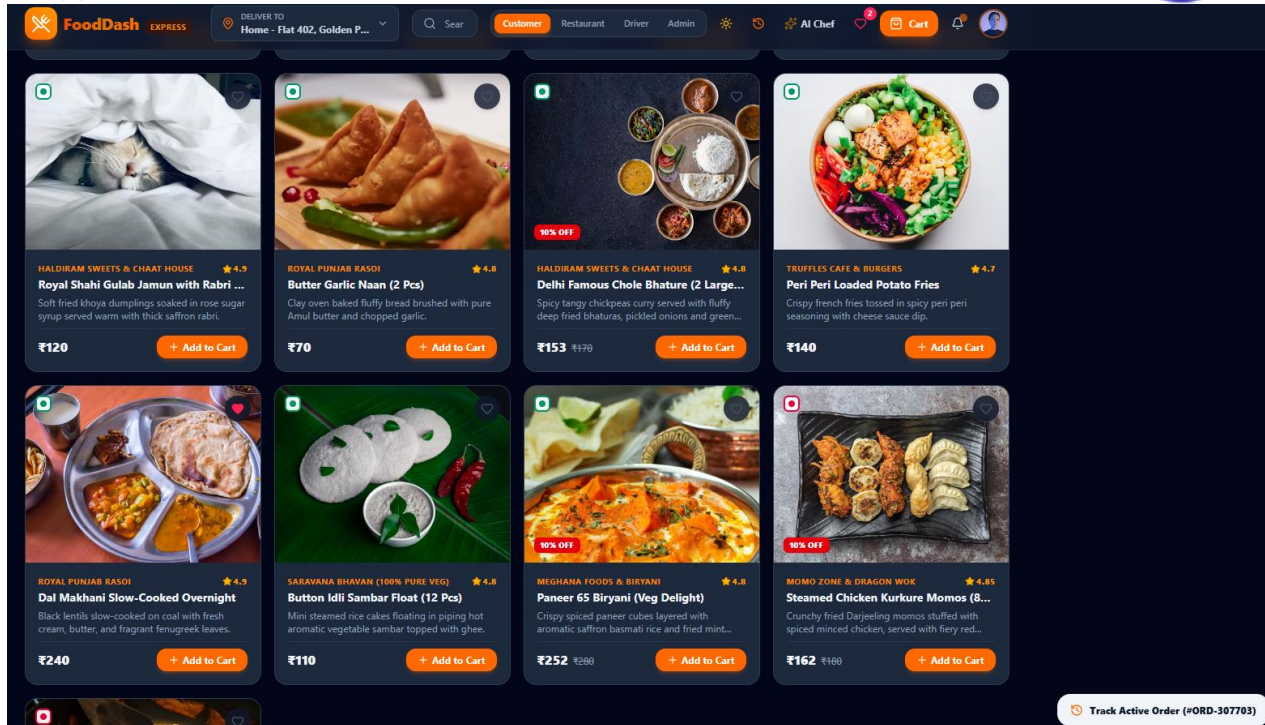
6.3 Performance Outcome

Parameter	Result
Home Page Load Time	~1.3 sec
Restaurant API Response	~180 ms
Food API Response	~150 ms
Order API Response	~240 ms
AI Recommendation Response	~3.2 sec
Memory Usage	~110 MB
CPU Usage	Below 35% during normal usage
Concurrent Users Tested	25 Users
Success Rate	100%
Application Availability	Stable

The FoodDash-Express application demonstrated stable performance under normal operating conditions. All REST APIs responded within acceptable limits, order processing was completed efficiently, and the Rasoi AI assistant generated recommendations within a few seconds. The system maintained low memory and CPU usage, indicating that the architecture is suitable for small to medium-scale food delivery applications and can be further scaled by integrating a persistent

6.4 Project Snapshots





FoodDash EXPRESS

Customer Restaurant **Driver** Admin

DELIVERY PARTNER
Ramesh Gowda
 Vehicle: KA-01-ET-4821 • Rating: ★ 4.9

Online for Deliveries ☒

Today's Earnings **₹220**

Completed Trips **1242**

Status **Available for Orders**

No active trips assigned
 Keep status Online to receive incoming food delivery requests.

FoodDash EXPRESS

Customer Restaurant Driver **Admin**

SYSTEM COMMAND CENTER
Admin Analytics & Management

+ New Coupon + New Category

Total Platform Revenue **₹7330**

Total Orders **189**

Registered Users **4**

Restaurants Partnered **6**

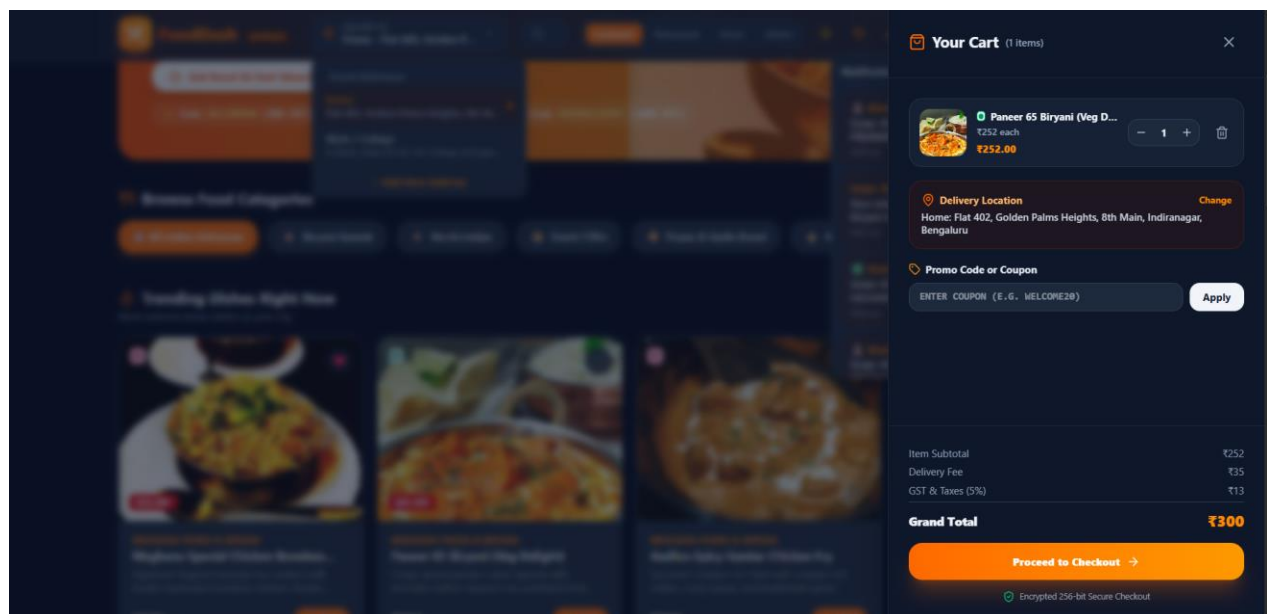
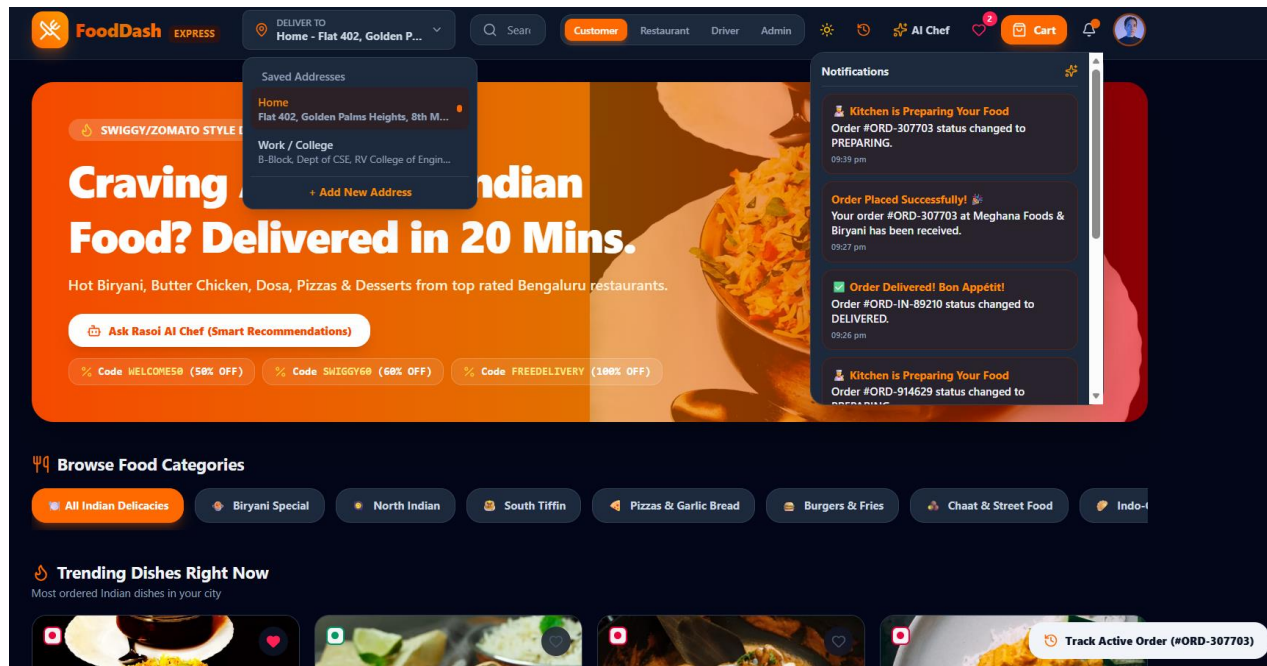
Analytics Charts Users Management Restaurants (6) Coupons & Discounts (3)

Weekly Revenue Trend (₹)

Day	Revenue (₹)
Mon	1500
Tue	2000
Wed	2500
Thu	2000
Fri	3500
Sat	4500
Sun	4000

Top Selling Dishes (Total Orders)

Dish	Total Orders
Margherita Pizza	350
Chicken Biryani	450
Smash Burger	300
Tonkotsu Ramen	200
Tacos Trio	150



Checkout & Payment

Delivering To: Home

Flat 402, Golden Palms Heights, 8th Main, Indiranagar, Bengaluru - 560038

Select Payment Method

Card

Instant UPI

Cash (COD)

CARD NUMBER

4532 **** * 8921

EXPIRY (MM/YY) CVV / CVC

12/28 ...

Total Payable

₹300

Pay & Place Order

FoodDash
EXPRESS

DELIVER TO
Home - Flat 402, Golden P...

Search

Customer
Restaurant
Driver
Admin

AI Chef

Cart

Order Received
Sent to restaurant kitchen

Preparing Food
Chef is cooking your dish

Ready for Pickup
Food packed & waiting for...

Out for Delivery
Driver is on the way to you

Delivered
Enjoy your delicious meal!

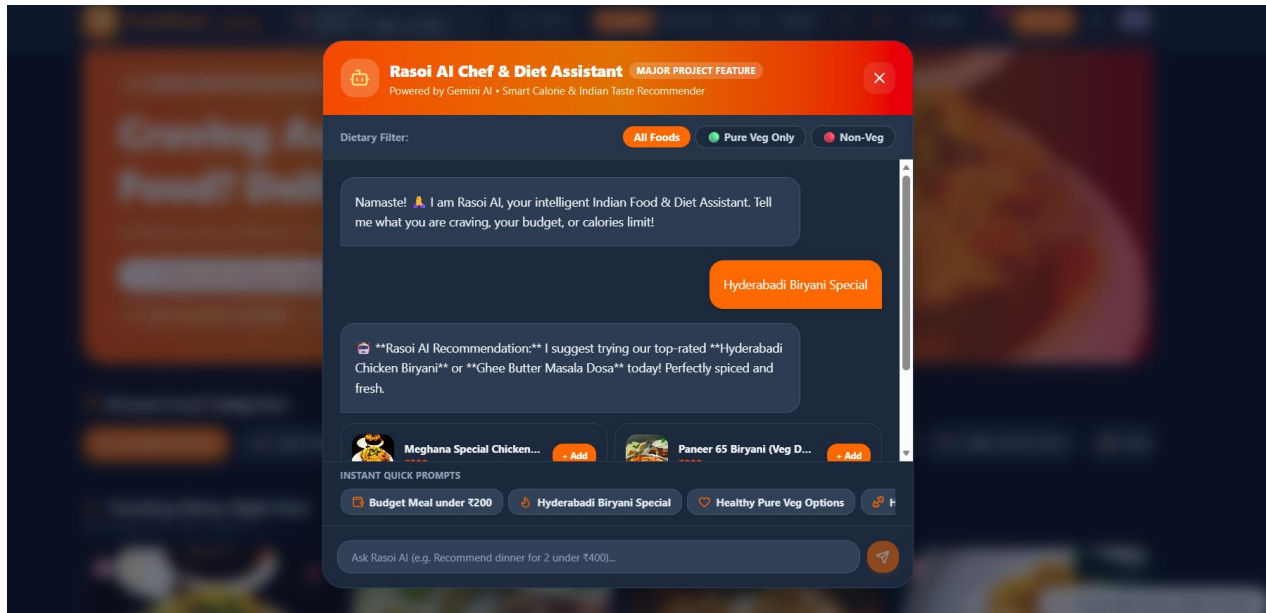
Live Delivery GPS Route

Meghana Foods & Biryani

Delivery Destination

David Miller
Vehicle: NY-TX-9021 • 4.9

Call Delivery Partner



7. My learnings

The development of the FoodDash-Express project provided me with valuable hands-on experience in designing and building a full-stack web application. Throughout this project, I enhanced my knowledge of React 19, TypeScript, Node.js, Express.js, RESTful APIs, Tailwind CSS, and Vite while understanding how frontend and backend components work together to create a complete application.

I learned how to design a modular and scalable system architecture, implement CRUD operations, manage application state, and develop role-based portals for customers, restaurants, delivery partners, and administrators. Working on features such as order management, payment simulation, live delivery tracking, and AI-powered food recommendations improved my problem-solving and logical thinking skills.

This project also strengthened my understanding of software development best practices, including responsive UI design, API integration, debugging, testing, performance optimization, and version control. Integrating the Google Gemini AI assistant gave me practical exposure to incorporating AI services into real-world applications.

Overall, this project significantly improved my technical knowledge, coding standards, teamwork, and project management skills. The experience has increased my confidence in developing modern full-stack web applications and prepared me for future opportunities in software development, web engineering, and AI-enabled application development.

8. Future work scope

The FoodDash-Express application has been developed as a functional full-stack food delivery platform. However, several enhancements can be implemented in the future to improve its functionality, scalability, and user experience.

Future improvements include integrating a real database such as MongoDB or PostgreSQL instead of using in-memory storage to ensure persistent data management. Secure user authentication and authorization can be implemented using JWT, OAuth, or Firebase Authentication to enhance system security.

The payment module can be upgraded by integrating real payment gateways such as Razorpay, Stripe, or PayPal for secure online transactions. The live delivery tracking feature can be improved using Google Maps API and real-time GPS tracking instead of simulation.

The AI-powered Rasoi AI Chef Assistant can be enhanced to provide personalized meal recommendations based on user preferences, order history, dietary restrictions, and nutritional information. Push notifications, email alerts, and SMS notifications can also be integrated to keep users informed about order updates.

For improved scalability and deployment, the application can be hosted on cloud platforms such as AWS, Microsoft Azure, or Google Cloud Platform, with Docker and Kubernetes for containerization and orchestration. Additional features such as multilingual support, voice-based ordering, loyalty and reward programs, customer reviews, restaurant ratings, and advanced analytics dashboards can further improve the platform.

Overall, these future enhancements would make the FoodDash-Express application more secure, scalable, intelligent, and suitable for real-world commercial deployment.

